Appendix D

Falsifiability Conditions for the Two Terminal Theorems

This appendix specifies, in precise terms, the empirical conditions under which the Concession Theorem and the Haitian Theorem would be *falsified*. The purpose is twofold. First, it demonstrates that both theorems are genuine empirical claims with specifiable defeat conditions. Second, it provides a reference standard against which critics can test proposed counter-examples, and against which future evidence can be assessed as the historical record develops past 2026.

I. Falsification Conditions for the Concession Theorem

The theorem's claim. For every non-kinetic reform R_i in the 1450–2026 dataset, $\Delta \max(R_i) = 0$: no non-kinetic reform has permanently reduced the Elite's extraction share as measured at the system apex.

What falsification requires. A genuine falsification of the Concession Theorem must satisfy *all three* of the following conditions simultaneously:

1. **Structural reduction in extraction share**: The reform must produce a measurable, permanent decline in the top-decile's share of total national wealth, with an absolute decrease in E's extraction share sustained over a minimum of three decades. A reduction in the growth rate of inequality falls short of the required structural reduction.

2. **No compensatory extraction elsewhere**: The reduction in the measured extraction share must not be offset by increased extraction through other mechanisms—capital flight to offshore jurisdictions, expansion of the extraction base to new populations, or substitution of financial extraction for labor extraction. The test is whether aggregate extraction by the apex (E) declined in real terms across all channels.

3. **Sustained across threat cycles**: The reduction must persist through subsequent periods of low kinetic threat, demonstrating structural alteration of the extraction

kernel. A gain that erodes once the threat subsides counts as evidence for the theorem.

The kinetic-precondition prediction. The theorem additionally predicts the *mechanism* by which concessions arise: reforms that reduce extraction load are offered when class resistance rises (min approaching τ), their magnitude scales with the threat level, and their durability decays as the threat retreats. This prediction is testable on its own terms; the Great Compression arc (rising with the kinetic convergence of 1932–1945, decaying after 1973 as the threat environment dissipated) is its cleanest measured instance (Figure 19.1). The presence of a kinetic precondition at adoption is evidence about the mechanism of concession; the three durability conditions above carry the falsification weight. A reform satisfying conditions 1–3 falsifies the theorem regardless of the threat conditions under which it was adopted.

The current state of the record. No reform in the 1450–2026 dataset satisfies all three conditions. The New Deal, the Civil Rights Act, OSHA, the Voting Rights Act, and comparable reforms in the dataset satisfy condition 1 partially and temporarily, but fail conditions 2 or 3 or both: the long-run wealth concentration data shows that $\Delta \max$ returned to zero within two to three decades in every case (Figure 19.1), and each reform's adoption under elevated min conditions matches the kinetic-precondition prediction. The theorem remains unfalsified.

What would falsify it. A concrete example: if post-2026 evidence showed that the top 10% wealth share in the United States had declined from approximately 70% (2020 baseline) to below 60% and remained there for thirty or more years following a non-kinetic reform (legislation, litigation, or electoral change), with no measurable compensatory extraction in other jurisdictions or asset classes, the Concession Theorem would require revision or scope restriction—whatever the threat environment at the moment of adoption.

II. Falsification Conditions for the Haitian Theorem

The theorem's claim. For all non-kinetic interventions R_i in the dataset: $\Delta \max(R_i) = 0$ (eq. 19.10). There exists at least one kinetic action K_j such that $\max(t_{\text{post}}) = 0$ locally (eq. 19.11).

What falsification requires. A genuine falsification of the strong-form Haitian Theorem must produce a documented historical case in which *all five* of the following conditions are satisfied:

1. **Intra-national racial partition**: The society in question must have an installed,

operative racial partition with a functioning ψ mechanism—a buffer class holding status wages against a racially defined out-group. Systems without this architecture fall under the theorem's scope condition (Appendix D, Section I) and cannot falsify the strong form.

2. **Absence of kinetic precondition**: The liberation must have been achieved without any credible kinetic threat operating in the background—no armed wing, no imminent insurrection, no defection within F_{enforce}, no external military force capable of enforcing a transition. If kinetic potential was present in any of these forms, the case resolves as the theorem's weak form (threat without discharge), which falls outside the falsification condition.

3. **Kernel-level liberation**: The outcome must terminate the prior extraction kernel; desegregation of public facilities while the economic commanding heights remain in Elite ownership is an interface update. Kernel termination requires that the prior Elite's extraction capacity was structurally reduced to zero in the material apparatus (the plantation, the corporation, the financial system, or whatever constitutes the extraction mechanism) independently of any legal or political face change.

4. **Durable outcome**: The liberation must persist through at least one full threat cycle without requiring new kinetic pressure to maintain it, which would demonstrate structural dismantlement of the kernel.

5. **No Elite class substitution**: The liberation must not have resulted in a new extraction kernel installed by a replacement Elite—the Zimbabwe pattern (Appendix to the Boundary Conditions, Chapter 19), in which kinetic liberation produces kernel transplant while leaving the extraction topology intact, does not falsify the theorem. Falsification requires permanent architectural dismantlement of the extraction kernel.

The current state of the record. No case in the 1450–2026 dataset satisfies all five conditions. The boundary cases examined in Chapter 19 (South Africa, Scandinavia, India, Zimbabwe) each fail at least one condition; each failure illuminates the structural architecture of the theorem. The theorem remains unfalsified in strong form.

What would falsify it. A concrete example: if a documented historical case could be established in which a society with an operative intra-national racial partition (comparable to the American 1705–2026 architecture) achieved $\max(t_{\text{post}}) = 0$ (the permanent dismantlement of the extraction kernel) through moral argument, electoral politics, litigation, or nonviolent civil disobedience, with no kinetic threat operating in any form, the Haitian Theorem's strong form would require revision.

III. Why These Conditions Have Not Been Met

The two falsification specifications above share a common structure: they require permanent, kernel-level structural change in a society with an installed racial partition. The framework explains the empty counter-example set as a consequence of the architecture itself: the extraction kernel is a self-reinforcing closed-loop system whose Lyapunov stability (Section 14.1.2) absorbs every perturbation into a descent direction of the system's energy function. On this account, a non-kinetic intervention is a perturbation within the system's invariant set, and only a force that operates outside the system's control architecture (one that severs the spanning-tree topology itself) can produce the structural change that falsification requires. The explanation is itself a hypothesis, held accountable by the conditions above: it fails on the day a counter-example meets them.

The theorems are falsifiable: specifiable evidence can count against them. They remain unfalsified because the architectural constraint that generates their predictions is still operative. The Concession Theorem and the Haitian Theorem would be falsified by the existence of a historical counter-example meeting the specifications above. No such counter-example has been identified in the 1450–2026 dataset. The burden of proof falls on the critic to supply one.